

MDCAT Entry Test Mock Series 2026

Mock Test 2 — Answer Key & Concept Review

180 MCQs · 180 Marks · Covers Enzymes, Evolution, Reproduction, Coordination & Control, Chemical Bonding, Equilibrium, Kinetics, Thermochemistry, Electrochemistry, Fluid Dynamics, Waves, Thermodynamics, Electrostatics, English & Logical Reasoning

Biology

- Q1. Enzymes are chemically:
✓ **Correct Answer: Proteins**
- Q2. Optimum pH of pepsin is:
✓ **Correct Answer: 2**
- Q3. Optimum temperature for human enzymes:
✓ **Correct Answer: 37°C**
- Q4. Lock & key model was proposed by:
✓ **Correct Answer: Emil Fischer**
- Q5. Induced fit model was proposed by:
✓ **Correct Answer: Koshland**
- Q6. Non-protein part of enzyme is called:
✓ **Correct Answer: Cofactor**
- Q7. If cofactor is organic, it is called:
✓ **Correct Answer: Coenzyme**
- Q8. Apoenzyme + Cofactor =
✓ **Correct Answer: Holoenzyme**
- Q9. Enzyme that works on starch:
✓ **Correct Answer: Amylase**
- Q10. Enzyme that works on protein in stomach:
✓ **Correct Answer: Pepsin**
- Q11. Trypsin is secreted by:
✓ **Correct Answer: Pancreas**
- Q12. Sucrase breaks sucrose into:
✓ **Correct Answer: Glucose + Fructose**
- Q13. Competitive inhibitor binds to:
✓ **Correct Answer: Active site**
- Q14. Example of competitive inhibitor:
✓ **Correct Answer: Malonate**
- Q15. Feedback inhibition occurs when product binds to:
✓ **Correct Answer: Allosteric site**
- Q16. Km value low means enzyme has:
✓ **Correct Answer: High affinity**

- Q17. At high temperature enzyme is:
✓ **Correct Answer: Denatured**
- Q18. Zymogen is:
✓ **Correct Answer: Inactive enzyme precursor**
- Q19. Penicillin is what type of inhibitor:
✓ **Correct Answer: Irreversible**
- Q20. Enzyme remains after reaction:
✓ **Correct Answer: Unchanged**
- Q21. pH changes enzyme activity by affecting:
✓ **Correct Answer: Ionization of active site**
- Q22. "Origin of Species" was written by:
✓ **Correct Answer: Darwin**
- Q23. Basic unit of evolution is:
✓ **Correct Answer: Population**
- Q24. Homologous organs have:
✓ **Correct Answer: Same structure diff function**
- Q25. Analogous organs have:
✓ **Correct Answer: Same function diff structure**
- Q26. Example of homologous organs:
✓ **Correct Answer: Human arm-Bat wing**
- Q27. Example of analogous organs:
✓ **Correct Answer: Bird wing-Insect wing**
- Q28. Vestigial organ in humans:
✓ **Correct Answer: Appendix**
- Q29. Lamarck's theory is:
✓ **Correct Answer: Inheritance of acquired characters**
- Q30. Genetic drift is more prominent in:
✓ **Correct Answer: Small population**
- Q31. Hardy-Weinberg equation for allele frequency:
✓ **Correct Answer: $p+q=1$**
- Q32. Darwin's finches were found in:
✓ **Correct Answer: Galapagos**
- Q33. Peppered moth color change is example of:
✓ **Correct Answer: Natural selection**
- Q34. Fossils are mostly found in:
✓ **Correct Answer: Sedimentary rock**
- Q35. Reproductive isolation leads to:
✓ **Correct Answer: New species**
- Q36. Miller-Urey experiment produced:
✓ **Correct Answer: Amino acids**

- Q37. Convergent evolution produces:
✓ **Correct Answer: Analogous organs**
- Q38. Allopatric speciation is due to:
✓ **Correct Answer: Geographical barrier**
- Q39. DNA hybridization shows:
✓ **Correct Answer: Relatedness of species**
- Q40. Wisdom teeth in humans are:
✓ **Correct Answer: Vestigial**
- Q41. Coevolution example:
✓ **Correct Answer: Flower-Bee (Both a & b)**
- Q42. Fertilization in humans occurs in:
✓ **Correct Answer: Fallopian tube**
- Q43. Testosterone is secreted by:
✓ **Correct Answer: Leydig cells**
- Q44. Spermatogenesis occurs in:
✓ **Correct Answer: Seminiferous tubules**
- Q45. Ovulation occurs on which day of 28-day cycle:
✓ **Correct Answer: Day 14**
- Q46. Corpus luteum secretes:
✓ **Correct Answer: Progesterone**
- Q47. Placenta does NOT:
✓ **Correct Answer: Produce sperm**
- Q48. Menopause stops:
✓ **Correct Answer: Both a & b**
- Q49. Acrosome contains:
✓ **Correct Answer: Hydrolytic enzymes**
- Q50. Human gestation period is:
✓ **Correct Answer: 9 months**
- Q51. Oogenesis produces how many functional eggs from one cell:
✓ **Correct Answer: 1**
- Q52. LH surge causes:
✓ **Correct Answer: Ovulation**
- Q53. Epididymis function is:
✓ **Correct Answer: Sperm storage & maturation**
- Q54. Implantation occurs in:
✓ **Correct Answer: Uterus wall**
- Q55. Menstrual phase involves shedding of:
✓ **Correct Answer: Endometrium**
- Q56. Sertoli cells function:
✓ **Correct Answer: Nourish developing sperms**

- Q57. Graafian follicle releases:
✓ **Correct Answer: Secondary oocyte**
- Q58. Primary sex organ in female:
✓ **Correct Answer: Ovary**
- Q59. Semen contains secretions from all EXCEPT:
✓ **Correct Answer: Kidney**
- Q60. After fertilization, zygote undergoes:
✓ **Correct Answer: Mitosis**
- Q61. Umbilical cord connects fetus to:
✓ **Correct Answer: Placenta**
- Q62. Part of brain that controls balance:
✓ **Correct Answer: Cerebellum**
- Q63. Insulin is secreted by:
✓ **Correct Answer: Beta cells**
- Q64. Reflex action is controlled by:
✓ **Correct Answer: Spinal cord**
- Q65. Resting membrane potential of neuron:
✓ **Correct Answer: -70mV**
- Q66. Saltatory conduction occurs in:
✓ **Correct Answer: Myelinated axon**
- Q67. Synaptic transmission is mainly:
✓ **Correct Answer: Chemical**
- Q68. Master gland of body:
✓ **Correct Answer: Pituitary**
- Q69. Insulin decreases blood:
✓ **Correct Answer: Glucose**
- Q70. ADH is secreted by:
✓ **Correct Answer: Posterior pituitary**
- Q71. ADH function:
✓ **Correct Answer: Increases water reabsorption**
- Q72. Sympathetic nervous system causes:
✓ **Correct Answer: Heart rate increase**
- Q73. Thyroxine is secreted by:
✓ **Correct Answer: Thyroid**
- Q74. Cerebrum controls:
✓ **Correct Answer: Thinking & memory**
- Q75. Medulla oblongata controls:
✓ **Correct Answer: Heartbeat & breathing**
- Q76. Myopia occurs when image forms:
✓ **Correct Answer: In front of retina**

Q77. Adrenaline is secreted by:

✓ **Correct Answer: Adrenal medulla**

Q78. Photoreceptors in human eye:

✓ **Correct Answer: Rods & Cones**

Q79. Hyperthyroidism causes:

✓ **Correct Answer: Weight loss**

Q80. Iodine is needed for which hormone:

✓ **Correct Answer: Thyroxine**

Q81. Thalamus acts as:

✓ **Correct Answer: Relay station for sensory info**

Chemistry

Q82. Which situation is most favorable for the formation of an ionic bond?

✓ **Correct Answer: One atom easily loses electrons while the other easily gains them.**

Q83. A compound conducts electricity only in molten or aqueous state but not in solid state. It is most likely:

✓ **Correct Answer: An ionic compound**

Q84. Why does sodium chloride have a high melting point?

✓ **Correct Answer: Strong electrostatic attraction between oppositely charged ions**

Q85. Shape of CH₄ according to VSEPR:

✓ **Correct Answer: Tetrahedral**

Q86. Bond angle in water H₂O is:

✓ **Correct Answer: 104.5°**

Q87. Which molecule is non-polar:

✓ **Correct Answer: CO₂**

Q88. Hydrogen bonding is strongest in:

✓ **Correct Answer: HF**

Q89. Sigma bond is formed by:

✓ **Correct Answer: Head-on overlap**

Q90. Pi bond is formed by:

✓ **Correct Answer: Sidewise overlap**

Q91. Which compound is least likely to conduct electricity under normal conditions?

✓ **Correct Answer: CH₄ (methane)**

Q92. Dipole moment of CO₂ is:

✓ **Correct Answer: Zero**

Q93. Ionic character increases if electronegativity difference is:

✓ **Correct Answer: > 1.7**

Q94. Magnesium reacts with chlorine to form MgCl₂. Why does one magnesium atom combine with two chlorine atoms?

✓ **Correct Answer: Magnesium has two valence electrons to donate.**

Q95. In a dative bond, the shared pair of electrons is provided by:

✓ **Correct Answer: One atom only**

Q96. A student builds a simple galvanic cell using zinc and copper electrodes. Initially, a bulb connected to the circuit glows brightly. After several hours, the bulb gradually becomes dim and eventually goes out. Why?

✓ **Correct Answer: The reactants have been consumed, reducing the cell's potential difference until the reaction can no longer produce current.**

Q97. Which property is expected if the force between particles in a compound is very strong?

✓ **Correct Answer: High melting point**

Q98. Standard enthalpy of formation of O₂ is:

✓ **Correct Answer: 0 kJ/mol**

Q99. A student accidentally connects the wires of a galvanic cell in reverse. The voltmeter now shows a negative reading instead of a positive one. What is the most reasonable conclusion?

✓ **Correct Answer: The direction of electron flow has been reversed because the terminals were interchanged.**

Q100 Bomb calorimeter measures:

✓ **Correct Answer: ΔE at constant V**

Q101 Enthalpy of neutralization of strong acid & strong base is always:

✓ **Correct Answer: -57.1 kJ/mol**

Q102 Which is state function:

✓ **Correct Answer: Enthalpy**

Q103 $\Delta H = \Delta E + P\Delta V$ is valid for:

✓ **Correct Answer: Constant P**

Q104 A chemistry teacher places solid sodium chloride (NaCl) in an electric circuit, but the bulb does not glow. She then melts the same NaCl, and now the bulb lights up immediately. What best explains this?

✓ **Correct Answer: Ions in solid NaCl are fixed, while molten NaCl has free-moving ions.**

Q105 A reaction releases heat (exothermic). The reaction mixture is heated. How will the equilibrium respond?

✓ **Correct Answer: Shift toward reactants to absorb the added heat.**

Q106 For reaction $N_2 + 3H_2 \rightleftharpoons 2NH_3$, $K_c =$

✓ **Correct Answer: $\frac{[NH_3]^2}{[N_2][H_2]^3}$**

Q107 A catalyst is added to a system already at equilibrium. Which statement best describes the effect?

✓ **Correct Answer: The catalyst speeds up both forward and reverse reactions equally, so the equilibrium position remains unchanged.**

Q108 Catalyst affects equilibrium:

✓ **Correct Answer: No effect on position**

Q109 A cold drink bottle is sealed at the factory under high pressure. When the cap is opened, bubbles rapidly escape. Which explanation best describes this observation?

✓ **Correct Answer: Lowering pressure shifts the equilibrium so dissolved CO₂ escapes as gas.**

Q110 Two identical reversible reactions are carried out. In the second experiment, only a catalyst is added. After a long time, both experiments produce exactly the same amount of product, but the second reaction reaches it faster. Why?

✓ **Correct Answer: The catalyst only increases the rate at which equilibrium is reached.**

Q111 Increase in pressure favors side with:

✓ **Correct Answer: Less moles**

Q112 For exothermic reaction, K_p decreases with:

✓ **Correct Answer: Increase T**

Q113 Unit of K_c for $2\text{HI} \rightleftharpoons \text{H}_2 + \text{I}_2$:

✓ **Correct Answer: No unit**

Q114 A laboratory technician is asked to prepare hydrogen gas for an experiment. He has two equal masses of calcium carbonate: one is a single marble chip, and the other is finely powdered. Why does the powdered form react faster?

✓ **Correct Answer: Grinding increases the surface area, leading to more effective collisions per second, while the chemical bonding within the compound remains unchanged.**

Q115 Ionic product of water K_w at 25°C:

✓ **Correct Answer: 1×10^{-14}**

Q116 Two students perform the same chemical reaction. Student A keeps the reaction flask in ice water, while Student B keeps it in warm water. Both use the same reactants and quantities. Why does Student B's reaction proceed faster?

✓ **Correct Answer: More particles possess enough energy to overcome the activation energy.**

Q117 Unit of rate of reaction:

✓ **Correct Answer: $\text{mol L}^{-1} \text{s}^{-1}$**

Q118 A factory introduces a catalyst into a reaction. The products are obtained much faster, but laboratory analysis shows the amount of product at completion remains unchanged. What is the reason?

✓ **Correct Answer: The catalyst provides an alternative pathway with lower activation energy.**

Q119 For zero order reaction, unit of k:

✓ **Correct Answer: $\text{mol L}^{-1} \text{s}^{-1}$**

Q120 A chemist reacts hydrochloric acid with two samples of calcium carbonate: one large marble chip and one sample crushed into powder. Both have the same mass. Which reacts faster?

✓ **Correct Answer: Powder sample will react faster**

Q121 In the above question, what affects the rate of reaction if we consider marble chip will react faster? Why does the marble chip show this behavior?

✓ **Correct Answer: Marble exposes more surface area for collisions.**

Q122 A chemical engineer wants to speed up production while using the same chemicals. He may increase temperature, add a catalyst, or increase concentration. Which of the following changes products fastest without altering the amount of product formed?

✓ **Correct Answer: Adding a catalyst**

Q123 A sealed container contains nitrogen dioxide (NO₂) and dinitrogen tetroxide (N₂O₄). After some time, the concentrations stop changing, but microscopic observation shows molecules are still reacting. What does this indicate?

✓ **Correct Answer: Forward and reverse reactions occur at equal rates.**

Q124 Catalyst increases rate by:

✓ **Correct Answer: Decreasing E_a**

Q125 A sealed vessel contains gases at equilibrium. Suddenly, more reactant gas is injected without changing temperature. What will happen immediately after this disturbance?

✓ **Correct Answer: The system shifts to consume some of the added reactant.**

Q126 Arrhenius equation: $k =$

✓ **Correct Answer: $Ae^{(-E_a/RT)}$**

Physics

Q127 Which is not a limitation of Bernoulli's equation?

✓ **Correct Answer: Valid for compressible fluid**

Q128 Unit of Kinetic Energy is same as?

✓ **Correct Answer: Work**

Q129 A perfume sprayer works on the principle of?

✓ **Correct Answer: Bernoulli's principle**

Q130 Which process increases viscosity of blood?

✓ **Correct Answer: Dehydration**

Q131 In streamline flow the net force on fluid is?

✓ **Correct Answer: Directed along streamline**

Q132 Which force causes backward drag on an object in fluid?

✓ **Correct Answer: Viscous force**

Q133 A nozzle reduces diameter of pipe from 6cm to 3cm. What is the change in velocity of fluid if initial velocity was 2m/s?

✓ **Correct Answer: 8m/s**

Q134 If radius is doubled, terminal velocity increases by?

✓ **Correct Answer: 4×**

Q135 A liquid jet exits at 10 m/s from 5 m height. Time to hit ground?

✓ **Correct Answer: 1.0 s**

Q136 Maximum velocity and maximum displacement for SHM are numerically equal when time period is?

✓ **Correct Answer: 2π**

Q137 The length of second pendulum on the surface of earth is about?

✓ **Correct Answer: 99 cm**

Q138 Mechanical waves on the surface of liquid are?

✓ **Correct Answer: Both A & B**

Q139 In wave motion the least distance between two points which are out of phase?

✓ **Correct Answer: $\lambda/2$**

Q140 Stars moving away from earth show?

✓ **Correct Answer: Increase in wavelength**

Q141 A string of 7 m length has a mass of 0.035 kg. If tension is 80.5 N then speed of wave is?

✓ **Correct Answer: 110 m/s**

Q142 If the length of simple pendulum is increased by 300%, then the time period will increase by?

✓ **Correct Answer: 100%**

Q143 Doppler's effect is applicable to?

✓ **Correct Answer: All of these**

Q144 Which thermodynamic law indicate quality as well as quantity of energy?

✓ **Correct Answer: 2nd Law of Thermodynamics**

Q145 If system losses heat of 300J by performing work of 400J on the system. What is the change in internal energy?

✓ **Correct Answer: 100J**

Q146 If 5 kg and 10 kg water are heated, ratio of heat capacities?

✓ **Correct Answer: 1:2**

Q147 In adiabatic expansion?

✓ **Correct Answer: $\Delta U = \text{Negative}$**

Q148 Acceleration of earth around sun is always?

✓ **Correct Answer: Radial**

Q149 In thermodynamics, Zeroth law is related with?

✓ **Correct Answer: Thermal equilibrium**

Q150 If the change in volume of a balloon is doubled as compared to the initial volume at constant temperature what will be the change in pressure?

✓ **Correct Answer: Halved**

Q151 Which law indicates the relationship shown in figure below?

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✓ **Correct Answer: Boyles Law**

Q152 If 110J heat is added to the system and 40J work is done, then amount of work done is?

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✓ **Correct Answer: 70 J**

Q153 Change in internal energy of ideal gas depends on?

•

✓ **Correct Answer: Change in temperature**

Q154 A hollow metal sphere of radius 5 cm is charged so that the potential on its surface is 10 V. The potential at the centre of the sphere is

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✓ **Correct Answer: 10 V**

Q155 A tiny positive test charge is placed at point P near a charged object. The test charge experiences a force of 4 N. The test charge is then replaced by another positive charge having twice the magnitude. Assuming the same position, what happens to the electric field at point P?

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✓ **Correct Answer: It remains unchanged**

Q156 A charged capacitor is disconnected from the battery. A student carefully increases the distance between its plates without touching the charge stored on it. What happens to the potential difference across the capacitor?

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✓ **Correct Answer: It increases**

Q157 According to Coulomb's Law, if both charges and radius are doubled, what is the effect on F?

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✓ **Correct Answer: Remains Same**

Q158 A point charge is placed outside a closed spherical Gaussian surface. No charge is enclosed by the surface. What is the net electric flux through the surface?

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✓ **Correct Answer: Zero**

Q159 Electron between plates separated by 20 mm and 1000 V. Force on electron?

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✓ **Correct Answer: 8×10^{-14} N**

Q160 Two electric field lines are drawn passing through a region of space. At one point, a student sketches them crossing each other. What does this imply?

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✓ **Correct Answer: Two different electric field directions exist at the same point (indicates an inconsistency, since field lines cannot actually cross)**

Q161 Two identical positively charged spheres repel each other with a force F. One sphere is touched with the ground and then brought back to its original position. What happens to the electrostatic force?

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✓ **Correct Answer: It becomes zero**

Q162 Two charges exert a force of 20 N in air. Without changing the charges or distance, they are completely immersed in a liquid having dielectric constant $K=4$. What will happen to the force?

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✓ **Correct Answer: 5 N**

English

Q163 Synonym of "Abort"

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✓ **Correct Answer: Cancel**

Q164 Analogy of "Poet : Poetry"

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✓ **Correct Answer: Teacher : Teaching**

Q165 A student studied consistently throughout the year. As a result, he performed exceptionally well in the final examination. The word "consistently" is closest in meaning to:

.

✓ **Correct Answer: Regularly**

Q166 Choose the sentence that is grammatically correct.

.

✓ **Correct Answer: Neither the teacher nor the students were present.**

Q167 Ali always leaves home early because traffic becomes heavier after 8:00 a.m. One day, he left at 8:30 a.m. and reached the office late. Which statement is best supported?

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✓ **Correct Answer: Heavy traffic likely delayed Ali.**

Q168 Identify the sentence with no grammatical error.

.

✓ **Correct Answer: They were discussing the problem seriously.**

Q169 The word "Reluctant" means:

.

✓ **Correct Answer: Hesitant**

Q170 The opposite of "Scarce" is:

.

✓ **Correct Answer: Plenty**

Q171 Despite working very hard, she _____ the competition.

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✓ **Correct Answer: lost**

Logical Reasoning

Q172 Pattern Recognition 2, 6, 12, 20, 30, ?

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✓ **Correct Answer: 42**

Q173 Four friends—Ali, Sara, Bilal, and Hina—are sitting in a row. Ali sits immediately left of Sara. Bilal is at one end. Hina is not beside Bilal. Who is sitting at the other end?

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✓ **Correct Answer: Sara**

Q174 After a factory installed air filters, the amount of smoke released decreased significantly. Which conclusion is most reasonable?

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✓ **Correct Answer: Air filters likely reduced smoke emissions.**

Q175 Odd One Out (Kilometer, Litre, Gram, Second — unit of distance vs. others)

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✓ **Correct Answer: Kilometer**

Q176 Odd one out (Doctor, Engineer, Teacher, Book — profession vs. object)

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✓ **Correct Answer: Doctor**

Q177 All doctors are educated. Some educated people are researchers. Which conclusion must be true?

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✓ **Correct Answer: Some doctors may be researchers.**

Q178 Ahmed walks 5 km north, then 5 km east, then 5 km south. Where is he relative to his starting point?

✓ **Correct Answer: 5 km East**

Q179 A visitor has a valid card and the correct PIN, but the fingerprint does not match. What happens?

✓ **Correct Answer: Access is granted.**

Q180 Monday, Wednesday, Friday, Sunday, ?

✓ **Correct Answer: Tuesday**